Problem Set 5

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1. Perform a Bayesian version of a correlation that you performed in last week's problem set: Use the dataset "seasonal workers." Compute the Bayesian version of a correlation for BMI and both blood pressure measures, then compare results to the Pearson and Spearman results you generated last week. This requires that you use a different package in R than last week's Bayesian questions because bayestestR package doesn't handle this well. Instead, to compute a Bayesian correlation test, you will need the BayesFactor package. You then compute the correlation using the correlationBF() function.

The Bayseian correlation of BMI and SBP yields a median of 0.32, similar to Pearson correlation coefficient, which was 0.35. The Bayes factor, 21.97, provides strong evidence against the null. For BMI and DBP, the Bayesian median is 0.42, similar to the Pearson correlation coefficient, which was 0.46. The Bayes factor, 671.96, provides very strong evidence against the null. Both Bayesian medians were less similar to the Spearman correlation coefficient than the Pearson correlation coefficient, but all correlation coefficients from both the Pearson and Spearman tests call within the 95% credible intervals from the Bayesian correlations.

```r
result_systolic = correlationBF(seasonal_workers$BMI, seasonal_workers$SBP)
result_diastolic = correlationBF(seasonal_workers$BMI, seasonal_workers$DBP)

describe_posterior(result_systolic)
```

```
## Summary of Posterior Distribution
##
## Parameter | Median |       95% CI |     pd |          ROPE | % in ROPE |    BF |         Prior
## ---------------------------------------------------------------------------------------------
## rho       |   0.32 | [0.12, 0.50] | 99.98% | [-0.05, 0.05] |        0% | 21.97 | Beta (3 +- 3)
```

```r
describe_posterior(result_diastolic)
```

```
## Summary of Posterior Distribution
##
## Parameter | Median |       95% CI |     pd |          ROPE | % in ROPE |     BF |         Prior
## ----------------------------------------------------------------------------------------------
## rho       |   0.42 | [0.21, 0.58] | 99.85% | [-0.05, 0.05] |        0% | 671.96 | Beta (3 +- 3)
```

2. Use the dataset called "women of two religions" and do the following regression analyses. For each regression analysis: first visualize the data, then construct the linear regression model, evaluate the residual plots, check linearity, check homoscedasticity, check independence and check normality of residuals.

```r
knitr::kable(
  describeBy(women_religion) %>% round(2)
)
```

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
Woman ID	1	100	50.50	29.01	50.5	50.50	37.06	1	100	99	0.00	-1.24	2.90
Current Age	2	100	34.28	6.41	34.0	34.45	6.67	18	46	28	-0.19	-0.59	0.64
fertility	3	100	8.45	2.92	9.0	8.49	2.97	1	15	14	-0.11	-0.41	0.29
height	4	100	158.97	9.15	159.0	159.21	8.90	135	176	41	-0.20	-0.50	0.91
weight	5	100	54.61	3.68	55.0	54.70	2.97	45	62	17	-0.19	-0.51	0.37
Religion	6	100	1.50	0.50	1.5	1.50	0.74	1	2	1	0.00	-2.02	0.05

(a) Children produced regressed on height

The regression model indicates a significant effect of height on fertility/children produced (Beta = 0.16, p < 0.001). Checking linearity by plotting the regression line shows a straight-line relationship between height and fertility. Plotting the residuals confirms homoscedasticity as there is a uniform distribution across height values. The Durbin-Watson test indicates potential autocorrelation (DW = 1.63, p = 0.03), but the DW plot is randomly scattered, suggesting independence. Finally, the residuals do not appear to be normally distributed based on the Q-Q plot, as well as the Shapriro-Wilk test (W = 0.96, p = 0.005). However, linear models are robust to violations of normality, and we can conclude there is an association between fertility and height based on these results.

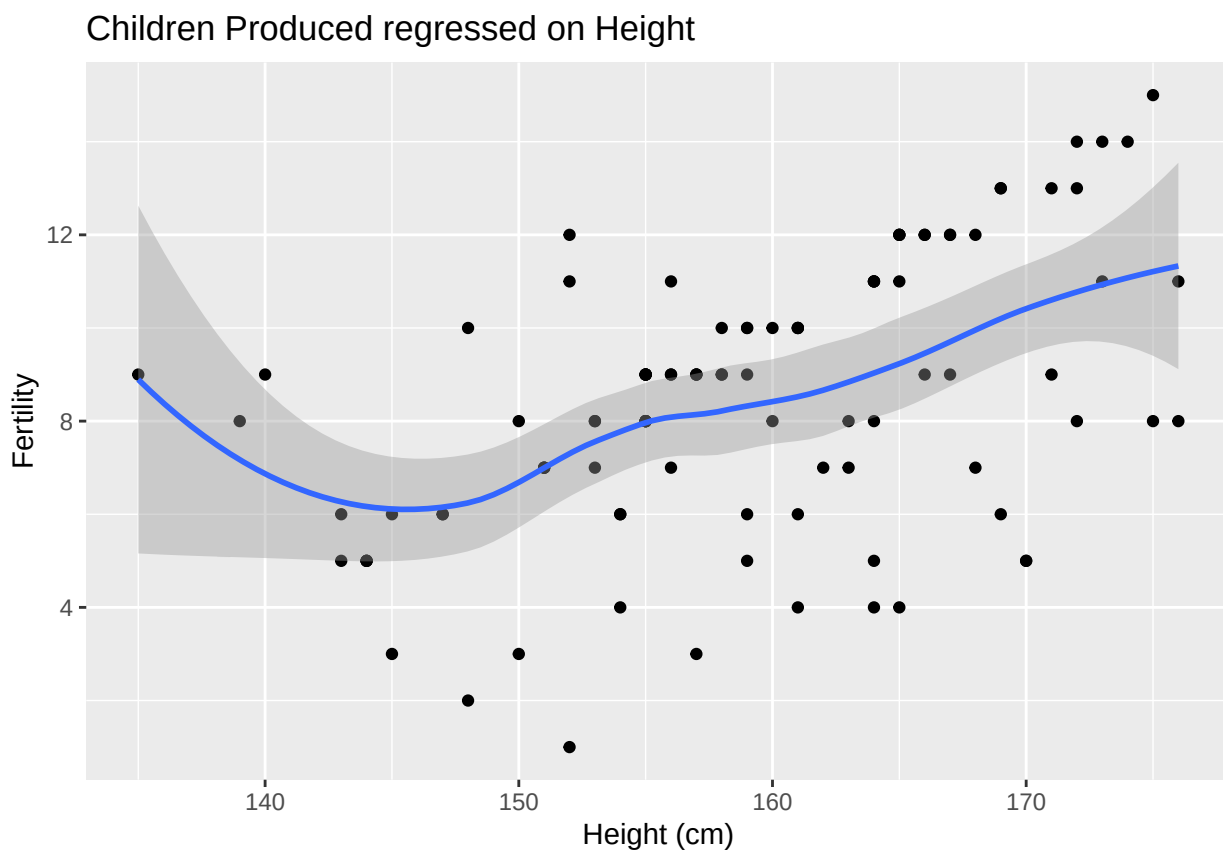
```r
#visualize the data
ggplot(women_religion, aes(x = height, y = fertility)) +
         geom_point() +
         stat_smooth() +
  labs(x = "Height (cm)", y = "Fertility", title = "Children Produced regressed on Height")
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

Linear regression model output (Fertility ~ Height)

Characteristic	Beta	95% CI	p-value
(Intercept)	-16	-25, -7.4	<0.001
height	0.16	0.10, 0.21	<0.001

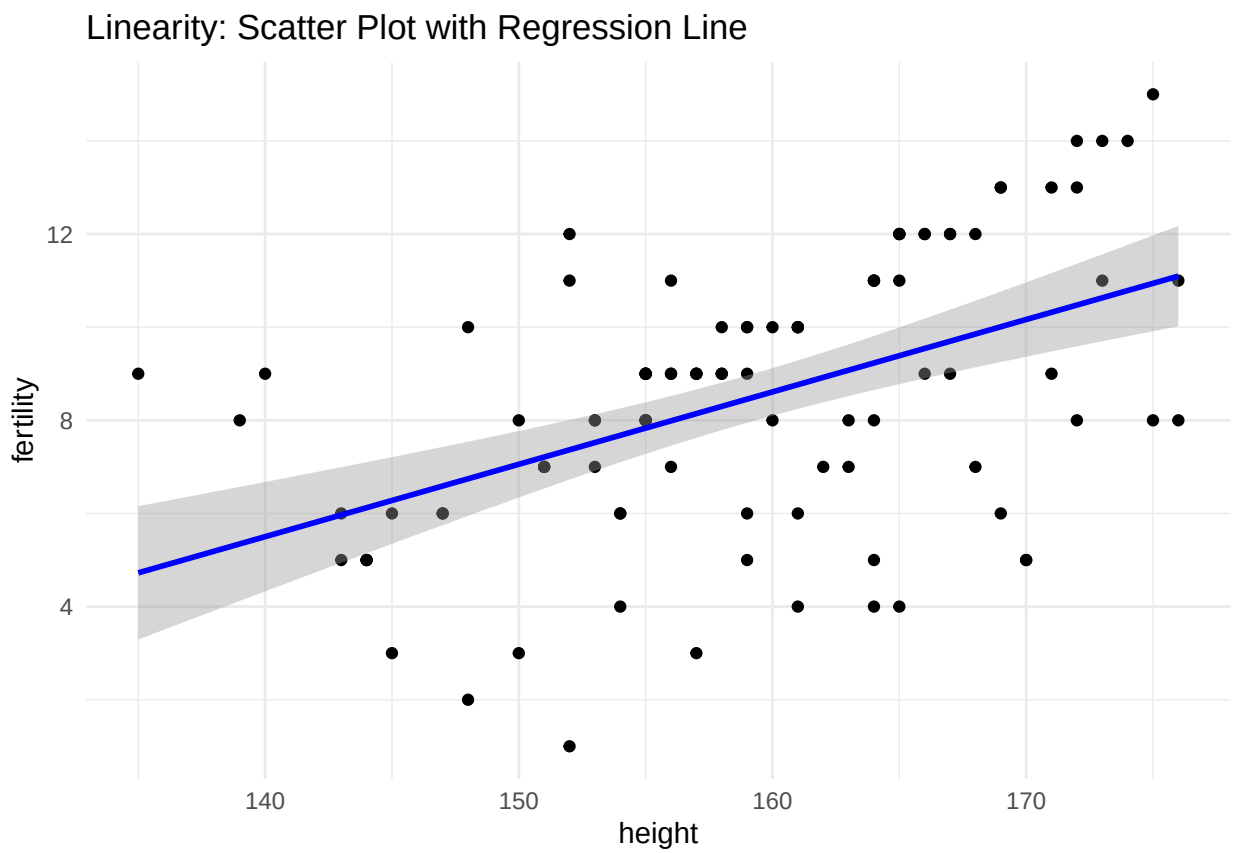
Abbreviation: CI = Confidence Interval



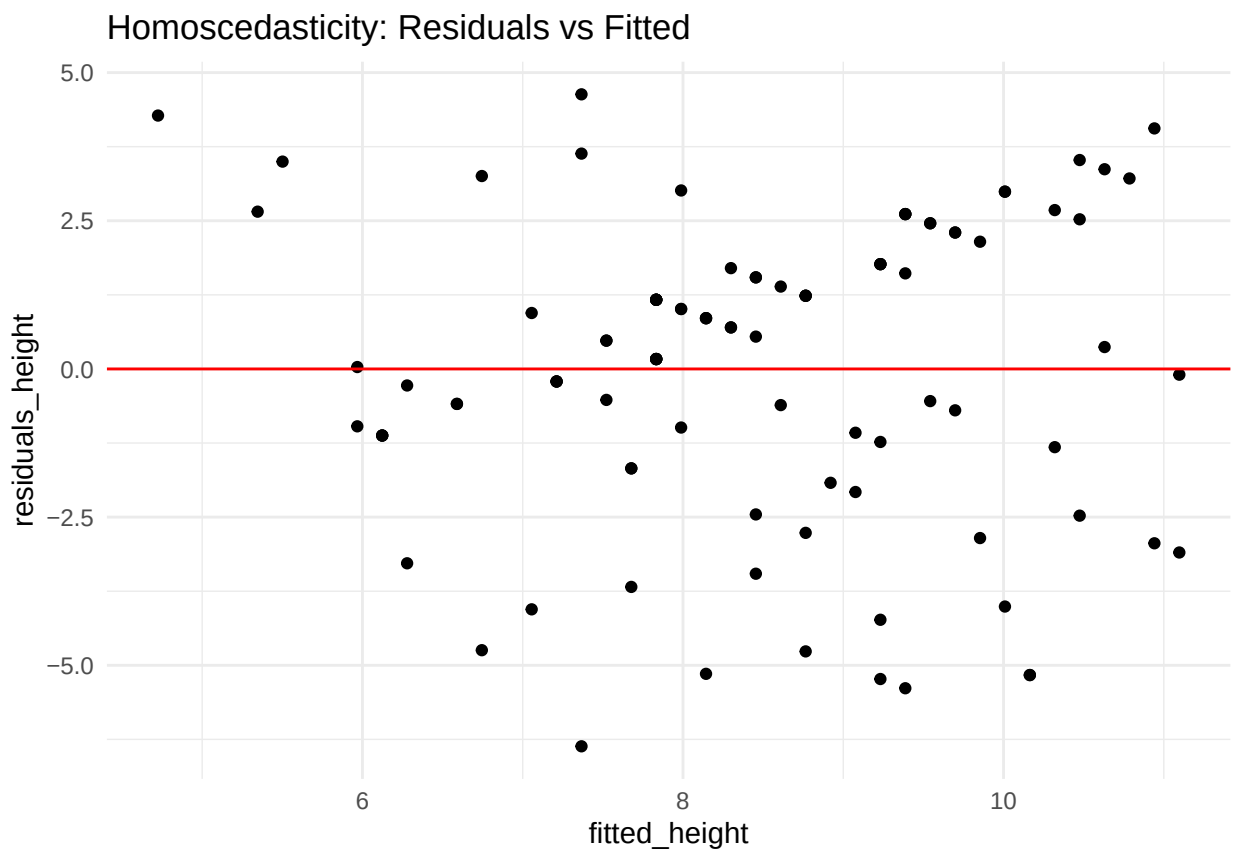

```
#construct model
model_height = lm(fertility ~ height, data = women_religion)
model_height_tbl <- tbl_regression(model_height, intercept = TRUE)
as_gt(model_height_tbl) |>
              tab_header("Linear regression model output (Fertility ~ Height)") |>
              tab_options(table.align='left')
```

```
#check linearity
ggplot(women_religion, aes(x = height, y = fertility)) +
  geom_point() +
  geom_smooth(method = "lm", color = "blue") +
  theme_minimal() +
  labs(title ="Linearity: Scatter Plot with Regression Line")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
#check homoscedasticity
residuals_height = resid(model_height)
fitted_height = fitted(model_height)
ggplot(women_religion, aes(x = fitted_height, y = residuals_height)) +
  geom_point() +
  geom_hline(yintercept = 0, color = "red") +
  theme_minimal() +
  labs(title = "Homoscedasticity: Residuals vs Fitted")
```

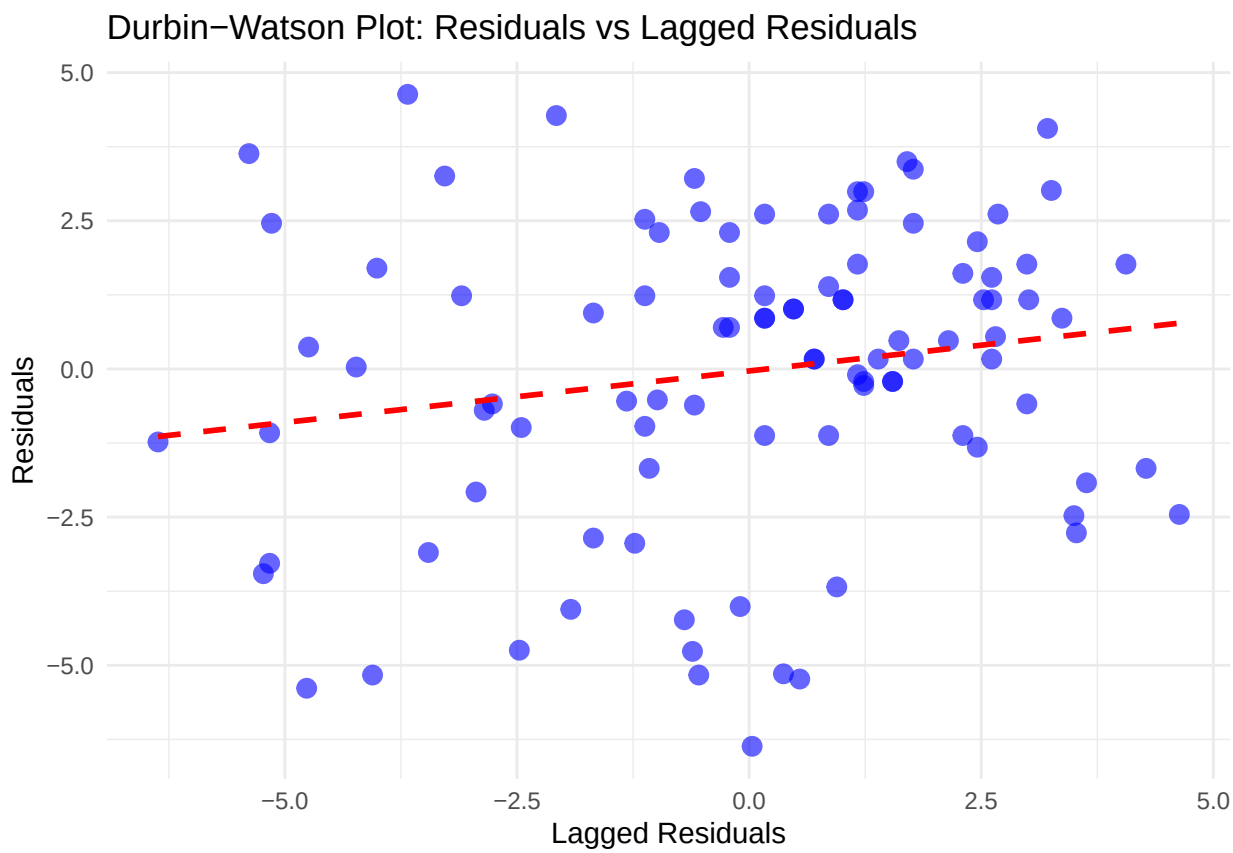



```r
#check independence
dwtest(model_height)
```

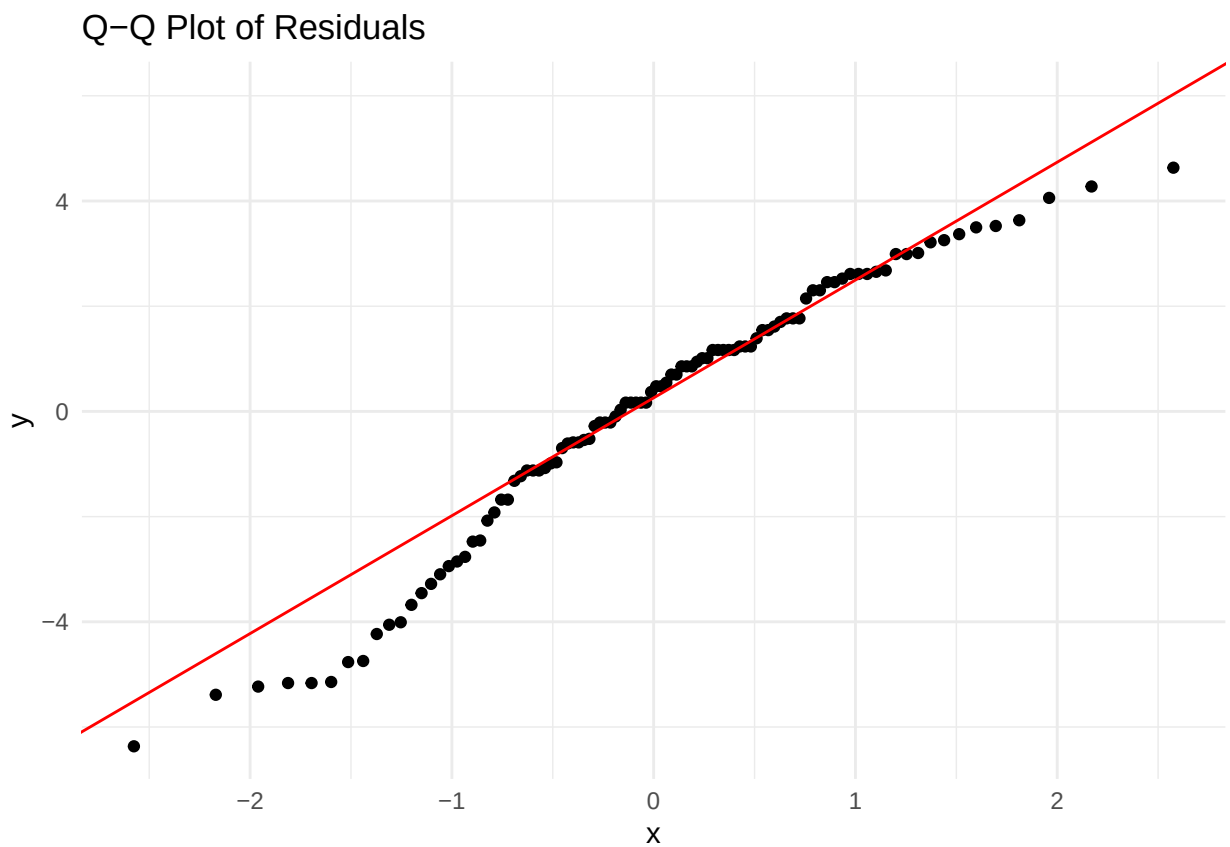
```
##
##  Durbin-Watson test
##
## data:  model_height
## DW = 1.6313, p-value = 0.03114
## alternative hypothesis: true autocorrelation is greater than 0
```

```r
residuals_height = resid(model_height)

dw_data_height = data.frame(
  residuals = residuals_height[-1],
  lagged_residuals = residuals_height[-length(residuals_height)]
)

ggplot(dw_data_height, aes(x = lagged_residuals, y = residuals)) +
  geom_point(color = "blue", size = 3, alpha = 0.6) +
  geom_smooth(method = "lm", color = "red", linetype = "dashed", se = FALSE) +
  theme_minimal() +
  ggtitle("Durbin-Watson Plot: Residuals vs Lagged Residuals") +
  xlab("Lagged Residuals") +
  ylab("Residuals")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
#check normality of residuals
ggplot(women_religion, aes(sample = residuals_height)) +
  geom_qq() +
  geom_qq_line(color = "red") +
  theme_minimal() +
  labs(title = "Q-Q Plot of Residuals")
```




```r
shapiro.test(model_height$residuals)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  model_height$residuals
## W = 0.96116, p-value = 0.004871
```

(b) Children produced regressed on weight

The regression model indicates a significant effect of weight on fertility/children produced (Beta = 0.38, p < 0.001). Checking linearity by plotting the regression line shows a straight-line relationship between weight and fertility. Plotting the residuals confirms homoscedasticity as there is a uniform distribution across weight values. The Durbin-Watson test indicates potential autocorrelation (DW = 1.65, p = 0.04), but the DW plot is randomly scattered, suggesting independence. Finally, the residuals do not appear to be normally distributed based on the Q-Q plot, as well as the Shapriro-Wilk test (W = 0.96, p = 0.003). However, linear models are robust to violations of normality, and we can conclude there is an association between fertility and weight based on these results.

```r
#visualize the data
ggplot(women_religion, aes(x = fertility, y = weight)) +
        geom_point() +
```

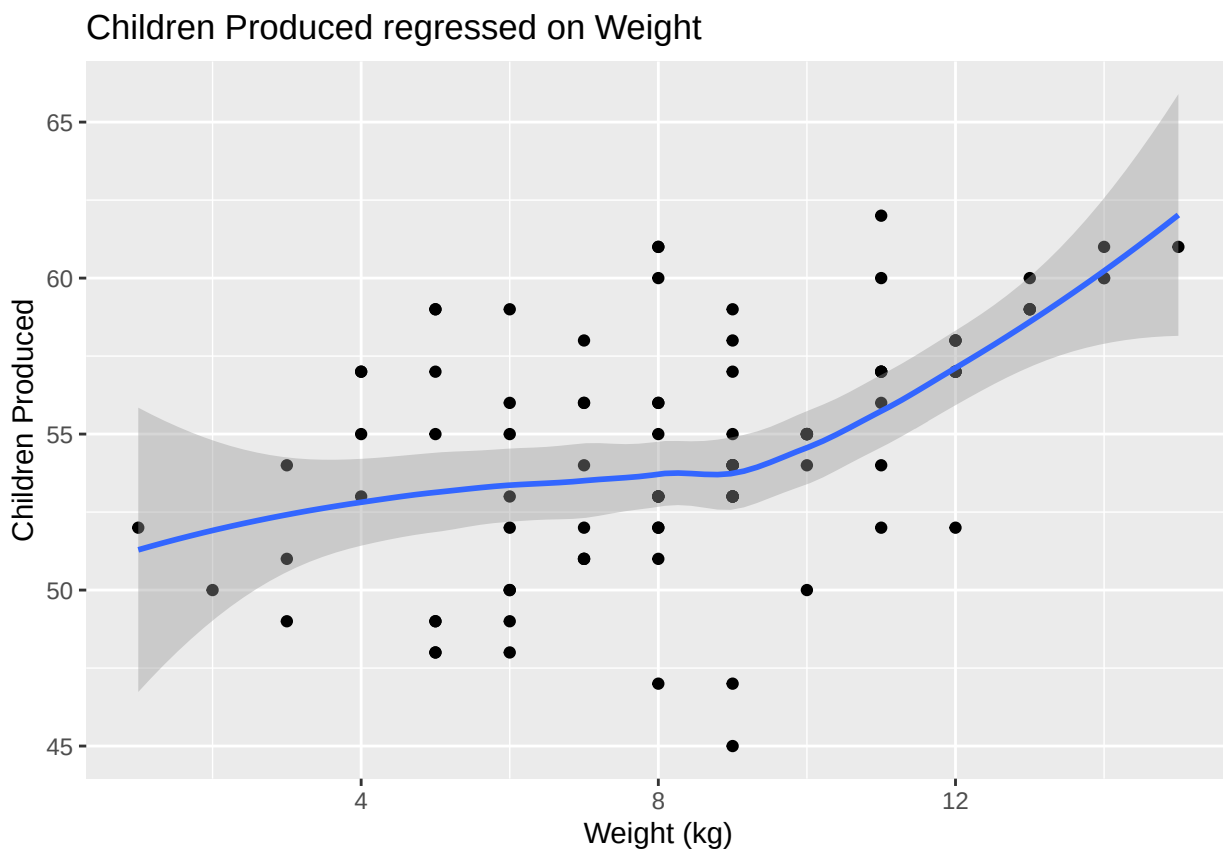
Linear regression model output (Fertility ~ Weight)

Characteristic	Beta	95% CI	p-value
(Intercept)	-13	-20, -4.9	0.002
weight	0.38	0.24, 0.52	<0.001

Abbreviation: CI = Confidence Interval

```
        stat_smooth() +
  labs(x = "Weight (kg)", y = "Children Produced", title = "Children Produced regressed on Weight")
```

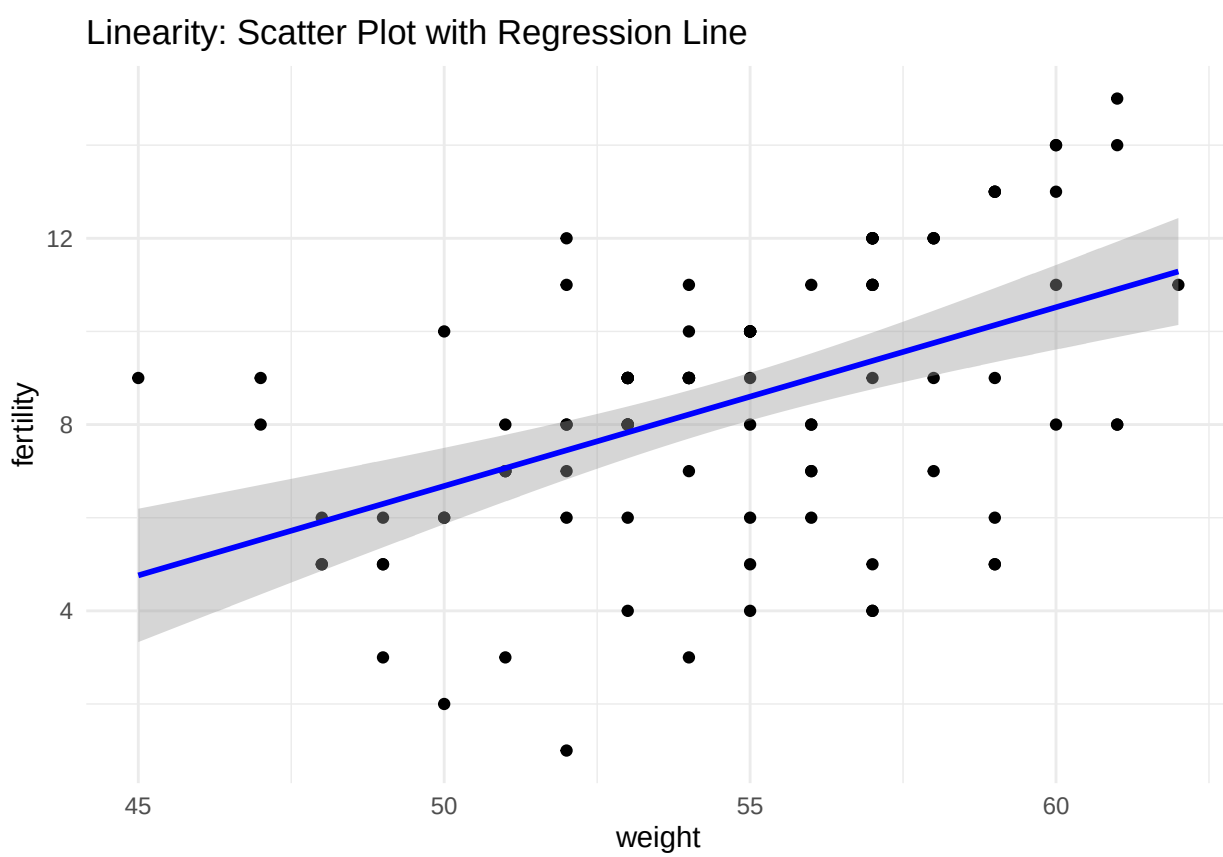
```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



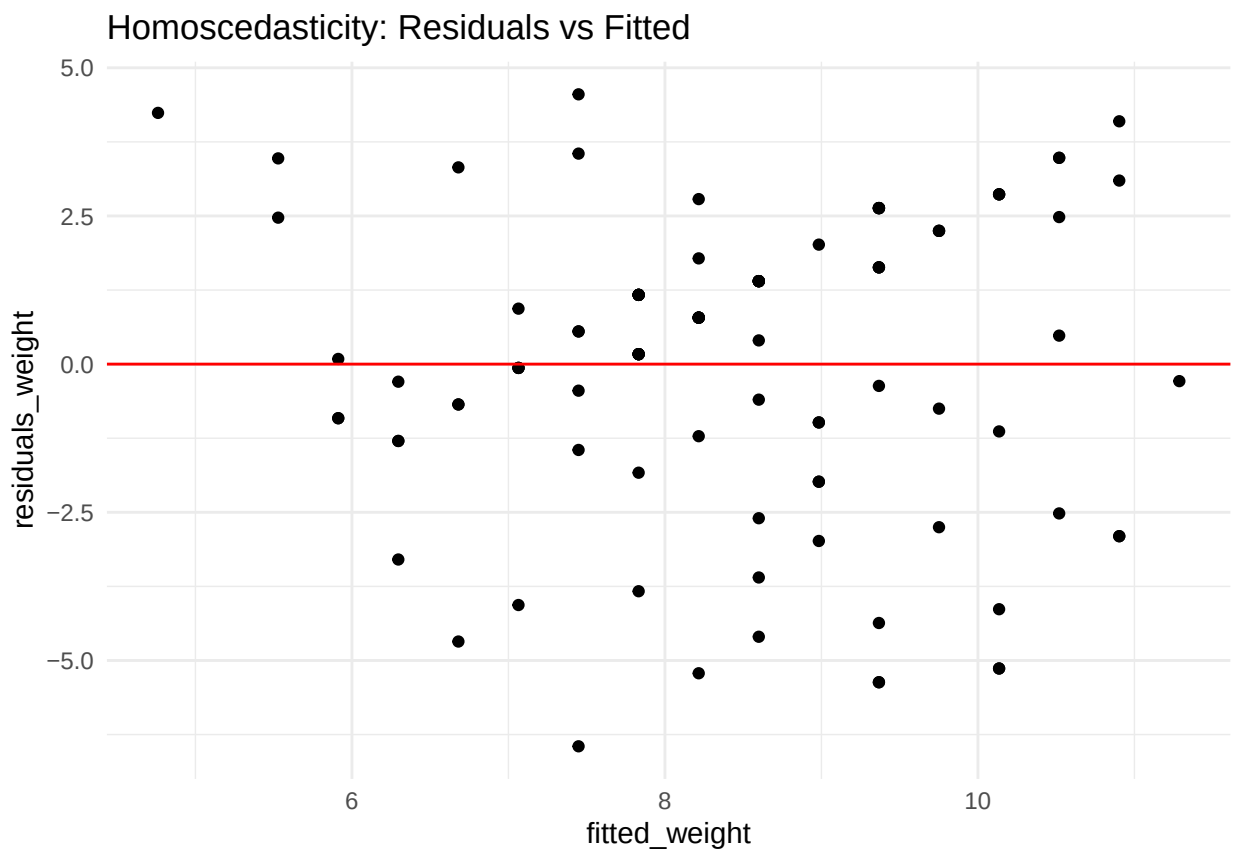

```
#construct model
model_weight = lm(fertility ~ weight, data = women_religion)
model_weight_tbl <- tbl_regression(model_weight, intercept = TRUE)
as_gt(model_weight_tbl) |>
              tab_header("Linear regression model output (Fertility ~ Weight)") |>
              tab_options(table.align='left')
```

```r
#check linearity
ggplot(women_religion, aes(x = weight, y = fertility)) +
  geom_point() +
  geom_smooth(method = "lm", color = "blue") +
  theme_minimal() +
  labs(title ="Linearity: Scatter Plot with Regression Line")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
#check homoscedasticity
residuals_weight = resid(model_weight)
fitted_weight = fitted(model_weight)
ggplot(women_religion, aes(x = fitted_weight, y = residuals_weight)) +
  geom_point() +
  geom_hline(yintercept = 0, color = "red") +
  theme_minimal() +
  labs(title = "Homoscedasticity: Residuals vs Fitted")
```

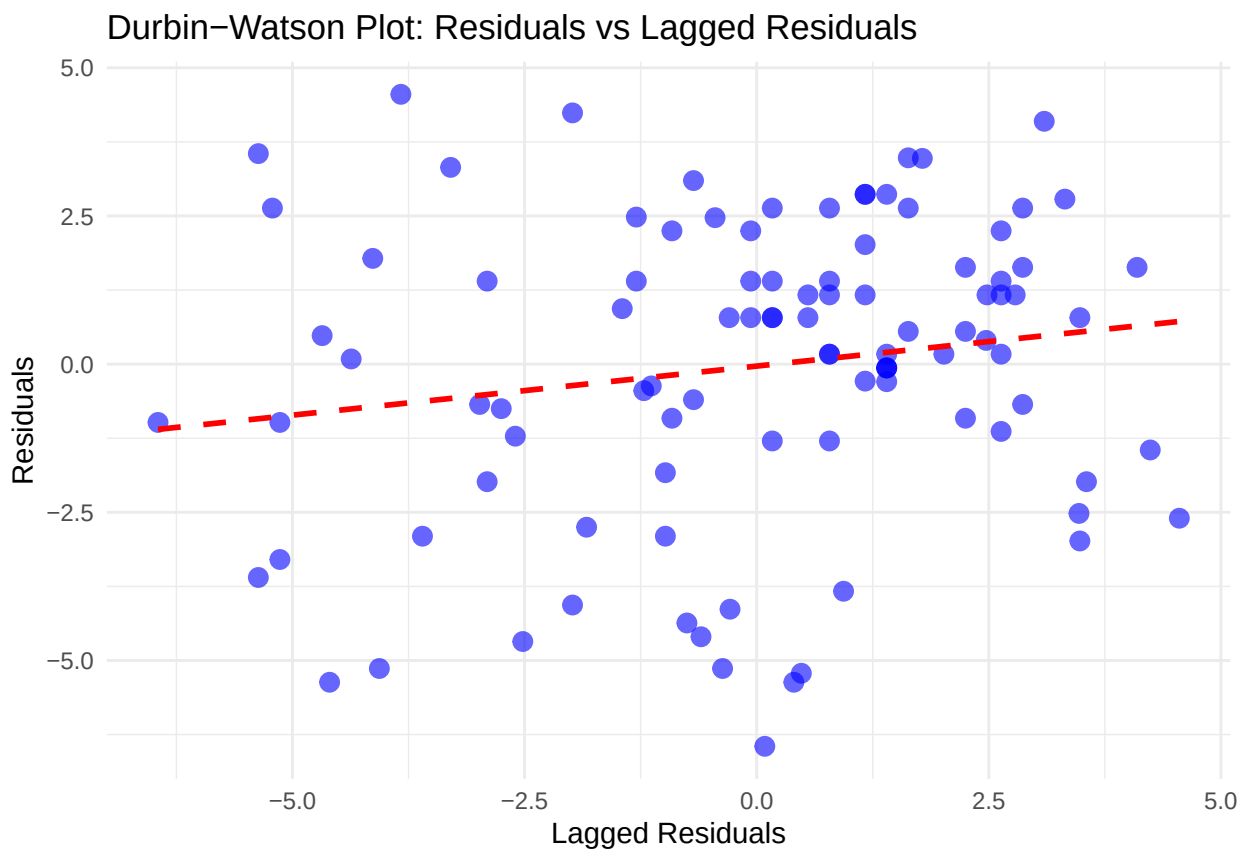



```r
#check independence
dwtest(model_weight)
```

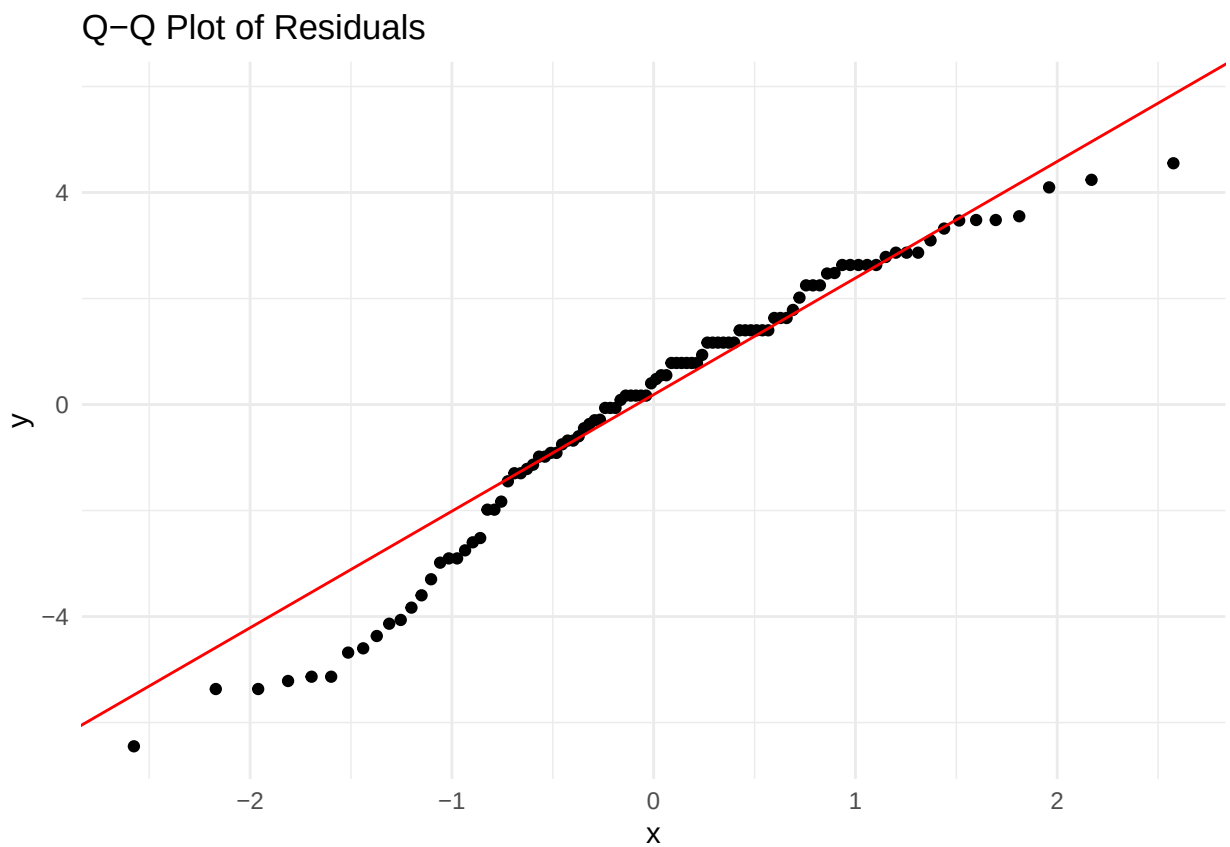
```
##
##  Durbin-Watson test
##
## data:  model_weight
## DW = 1.6487, p-value = 0.03779
## alternative hypothesis: true autocorrelation is greater than 0
```

```r
residuals_weight = resid(model_weight)

dw_data_weight = data.frame(
  residuals = residuals_weight[-1],
  lagged_residuals = residuals_weight[-length(residuals_weight)]
)

ggplot(dw_data_weight, aes(x = lagged_residuals, y = residuals)) +
  geom_point(color = "blue", size = 3, alpha = 0.6) +
  geom_smooth(method = "lm", color = "red", linetype = "dashed", se = FALSE) +
  theme_minimal() +
  ggtitle("Durbin-Watson Plot: Residuals vs Lagged Residuals") +
  xlab("Lagged Residuals") +
  ylab("Residuals")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
#check normality of residuals
ggplot(women_religion, aes(sample = residuals_weight)) +
  geom_qq() +
  geom_qq_line(color = "red") +
  theme_minimal() +
  labs(title = "Q-Q Plot of Residuals")
```




```
shapiro.test(model_weight$residuals)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  model_weight$residuals
## W = 0.95825, p-value = 0.003025
```

3. Use the dataset called "menarche" and regress number of children produced on age at menarche. First visualize the data, then construct the linear regression model, evaluate the residual plots, check linearity, check homoscedasticity, check independence and check normality of residuals.

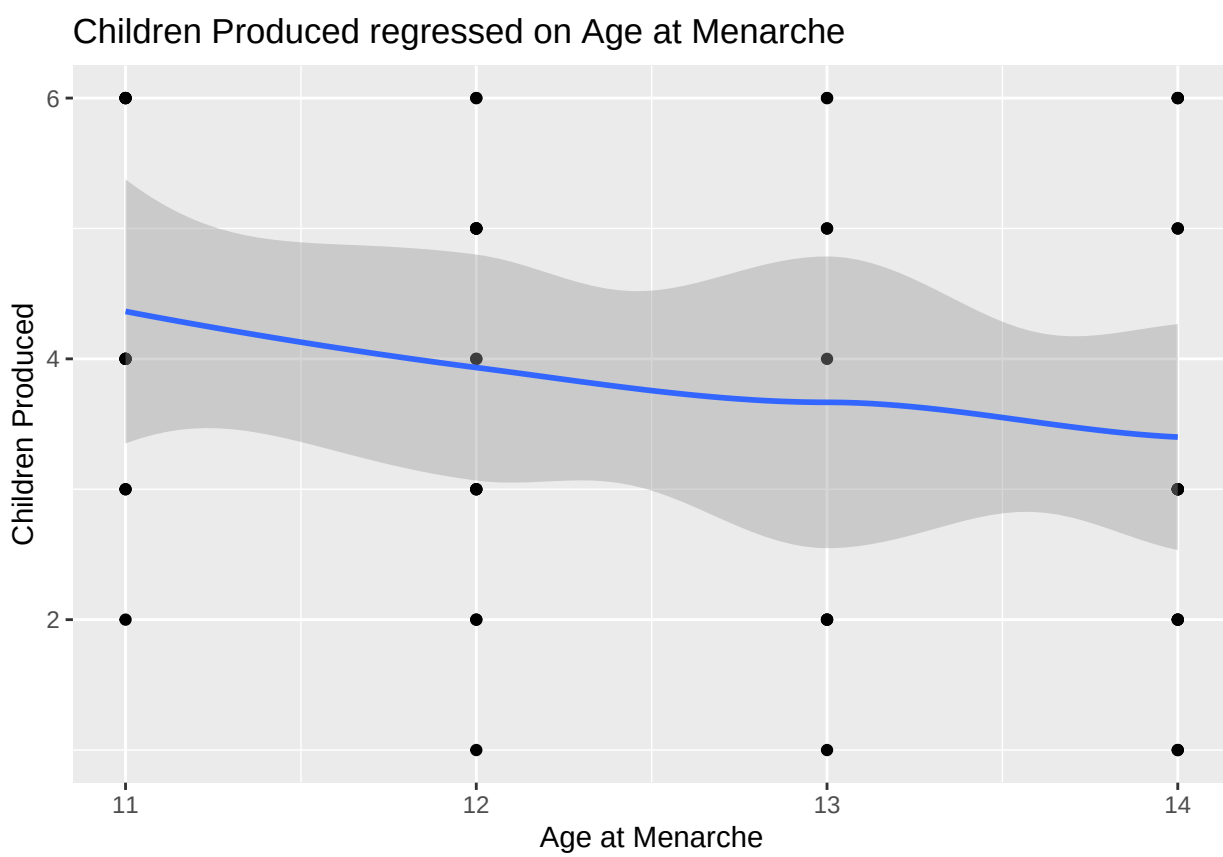
The regression model indicates a nonsignificant effect of age at menarche on fertility/children produced (Beta = -0.31, p = 0.13). Checking linearity by plotting the regression line shows a straight-line relationship between weight and fertility. Plotting the residuals confirms homoscedasticity as there is a uniform distribution across weight values. The Durbin-Watson test and plot indicate no autocorrelation (DW = 2.08, p = 0.6). Finally, the residuals do not appear to be normally distributed based on the Q-Q plot, as well as the Shapriro-Wilk test (W = 0.95, p = 0.03). However, linear models are robust to violations of normality, and we can conclude there is no association between fertility and age at menarche based on these results.

```
knitr::kable(
  describeBy(menarche) %>% round(2))
```

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
Woman ID	1	50	25.50	14.58	25.5	25.50	18.53	1	50	49	0.00	-1.27	2.06
menarche_age	2	50	12.56	1.15	12.0	12.57	1.48	11	14	3	0.02	-1.47	0.16
Age at First Pregnancy	3	50	19.48	2.36	19.5	19.48	3.71	16	23	7	-0.07	-1.42	0.33
fertility	4	50	3.82	1.65	4.0	3.88	1.48	1	6	5	-0.09	-1.32	0.23
Number of Surviving Children	5	50	2.54	1.53	2.0	2.45	1.48	0	6	6	0.45	-0.26	0.22

```
#visualize the data
ggplot(menarche, aes(x = menarche_age, y = fertility)) +
         geom_point() +
         stat_smooth() +
  labs(x = "Age at Menarche", y = "Children Produced", title = "Children Produced regressed on Age at Me
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
#construct model
model_menarche = lm(fertility ~ menarche_age, data = menarche)
model_menarche_tbl <- tbl_regression(model_menarche, intercept = TRUE)
as_gt(model_menarche_tbl) |>
              tab_header("Linear regression model output (Fertility ~ Age at Menarche)") |>
              tab_options(table.align='left')
```

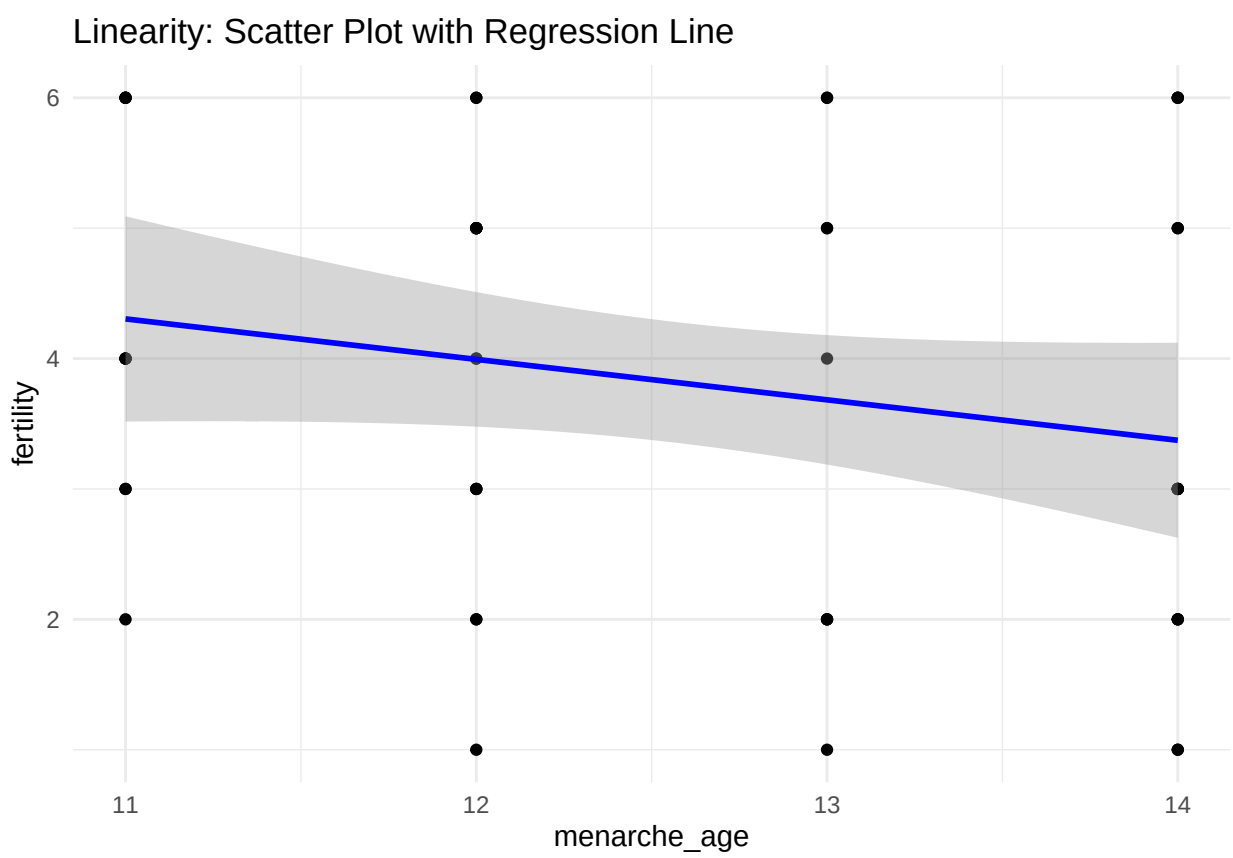
Linear regression model output (Fertility ~ Age at Menarche)

Characteristic	Beta	95% CI	p-value
(Intercept)	7.7	2.6, 13	0.004
menarche_age	-0.31	-0.72, 0.10	0.13

Abbreviation: CI = Confidence Interval

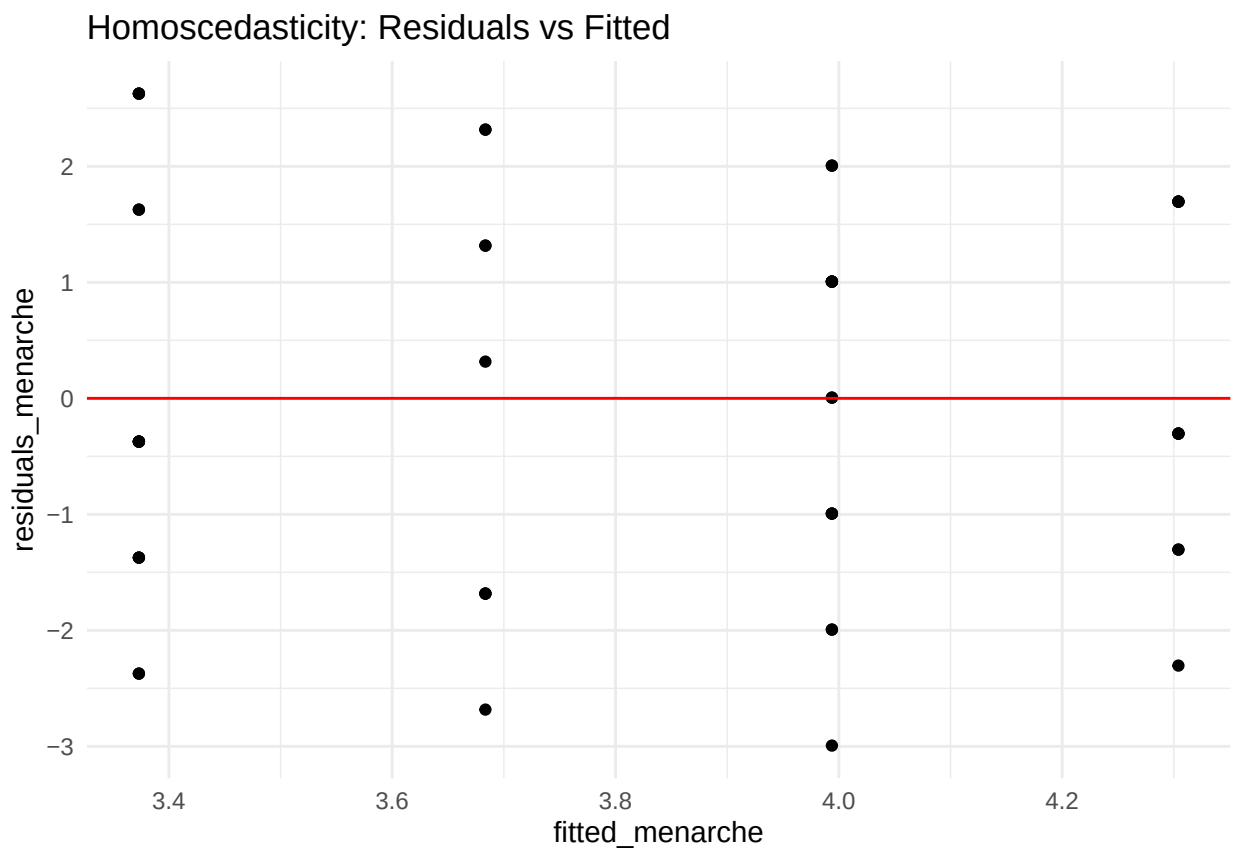
```
#check linearity
ggplot(menarche, aes(x = menarche_age, y = fertility)) +
  geom_point() +
  geom_smooth(method = "lm", color = "blue") +
  theme_minimal() +
  labs(title ="Linearity: Scatter Plot with Regression Line")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
#check homoscedasticity
residuals_menarche = resid(model_menarche)
fitted_menarche = fitted(model_menarche)
```

```r
ggplot(menarche, aes(x = fitted_menarche, y = residuals_menarche)) +
  geom_point() +
  geom_hline(yintercept = 0, color = "red") +
  theme_minimal() +
  labs(title = "Homoscedasticity: Residuals vs Fitted")
```



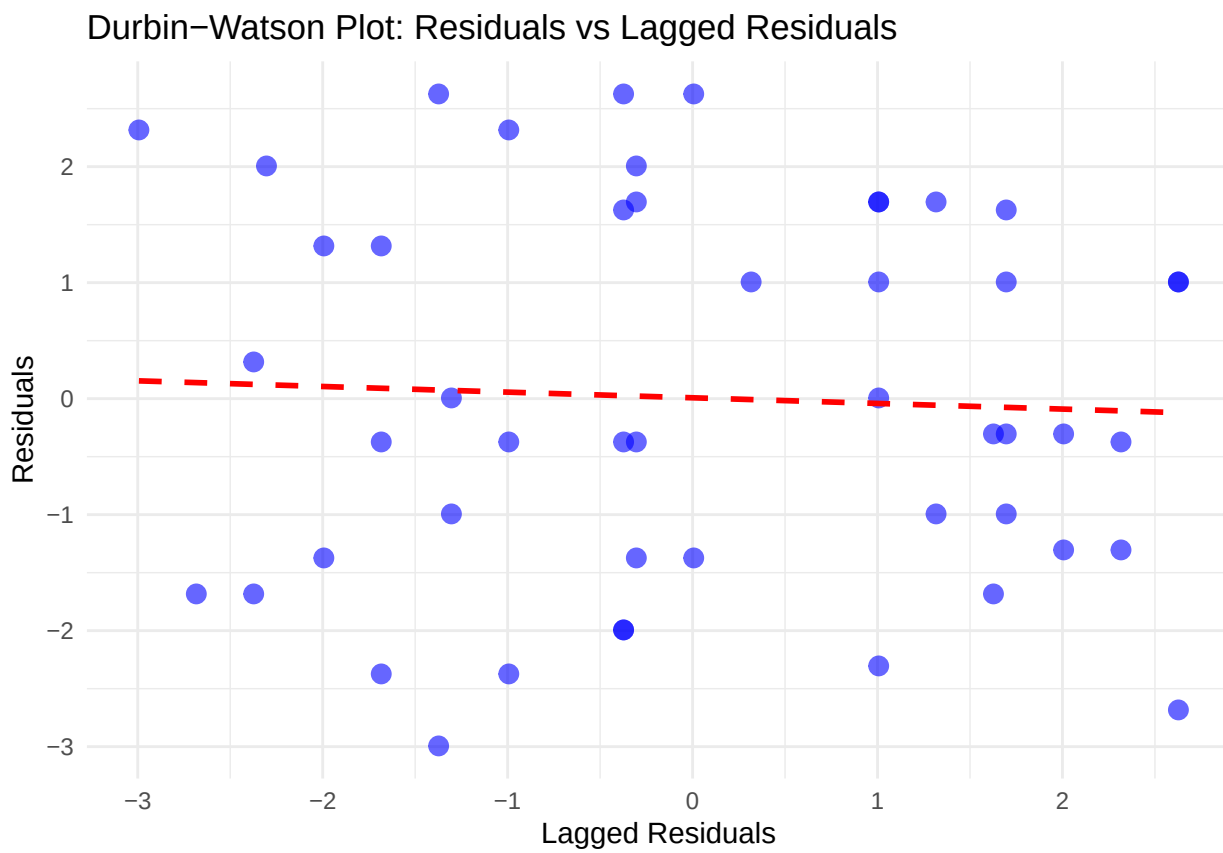

```r
#check independence
dwtest(model_menarche)
```

```
##
##  Durbin-Watson test
##
## data:  model_menarche
## DW = 2.0803, p-value = 0.6027
## alternative hypothesis: true autocorrelation is greater than 0
```

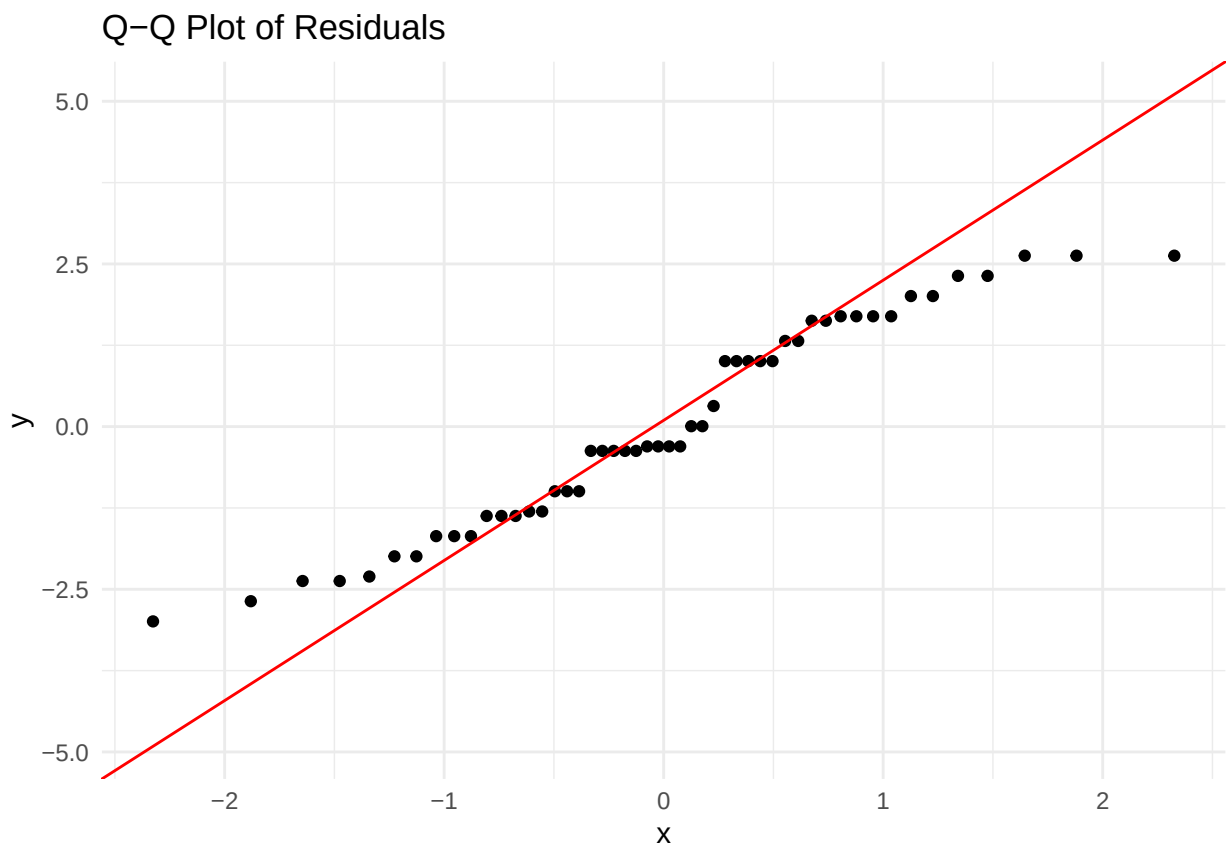
```r
residuals_menarche = resid(model_menarche)

dw_data_menarche = data.frame(
  residuals = residuals_menarche[-1],
  lagged_residuals = residuals_menarche[-length(residuals_menarche)]
)

ggplot(dw_data_menarche, aes(x = lagged_residuals, y = residuals)) +
```

```r
  geom_point(color = "blue", size = 3, alpha = 0.6) +
  geom_smooth(method = "lm", color = "red", linetype = "dashed", se = FALSE) +
  theme_minimal() +
  ggtitle("Durbin-Watson Plot: Residuals vs Lagged Residuals") +
  xlab("Lagged Residuals") +
  ylab("Residuals")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
#check normality of residuals
ggplot(menarche, aes(sample = residuals_menarche)) +
  geom_qq() +
  geom_qq_line(color = "red") +
  theme_minimal() +
  labs(title = "Q-Q Plot of Residuals")
```




```r
shapiro.test(model_menarche$residuals)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  model_menarche$residuals
## W = 0.94985, p-value = 0.03363
```